Middle Node

AlgoExpert · Easy · Linked Lists

The problem

- Return the middle node (second middle for even lengths) via slow/fast pointers.

Intuition

- A fast pointer moving two steps lands at the end exactly when a slow pointer moving one step reaches the middle.

Approach

- 1. Start slow and fast at the head
- 2. Advance slow by one, fast by two
- 3. Stop when fast runs out
- 4. Return slow

Java solution

```java
public class MiddleNode {
    public static LinkedListNode middleNode(LinkedListNode head) {
        LinkedListNode slow = head, fast = head;
        while (fast != null && fast.next != null) { slow = slow.next; fast = fast.next.next; }
        return slow;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- A fast pointer moving two steps lands at the end exactly when a slow pointer moving one step reaches the middle.
- Complexity: O(n)